This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2025.3643805

A Reliable Multi-Stage System for Tooth Instance Segmentation and Numbering in Panoramic Radiographs

Yani Zhang, Yuzhou Yu, Qiankun Li, *Member, IEEE*, Huabao Chen, Yuci Liang, Junxin Chen, *Senior Member, IEEE*, and Jiayue Yin

***Abstract*— Accurate tooth instance segmentation and identification in panoramic radiographs is a fundamental prerequisite for automated dental diagnosis and treatment planning. Existing methods often rely on static segmentation pipelines that overlook anatomical symmetry and fail to address structural anomalies such as supernumerary or missing teeth. To this end, we propose a reliable multi-stage framework for clinically applicable tooth instance segmentation and FDI-compliant numbering. The system comprises three stages: a supernumerary tooth classifier to guide adaptive routing, a core instance segmentor for tooth-level delineation and numbering, and a missing tooth detector to ensure index completeness via post-hoc correction. Central to our design is the Symmetry-Aware Dual-branch Pyramid Network (SADP-Net), which explicitly models bilateral structures and scale variations through a Symmetry-Aware Module (SymAM) and a Dual-Branch Pyramid (DBP) architecture. Extensive experiments on three datasets, including Tufts Dental, O^2PR, and a multi-pathology cohort, demonstrate superior performance compared to state-of-the-art baselines. Ablation studies further validate the contributions of each proposed component in enhancing boundary localization, numbering consistency, and robustness to anatomical variability. Our framework provides a scalable and interpretable solution for real-world dental imaging systems. Codes link: `https://github.com/qklee-lz/SADP-Net`.**

***Index Terms*— Tooth instance segmentation, Tooth numbering, Panoramic radiographs, Multi-stage system, Abnormal tooth adaptation, Symmetry-aware learning**

This work was supported by the Fundamental Research Funds for the Central Universities (No. DUT25YG229), and Xiaomi Young Talents Program.

Yani Zhang and Yuzhou Yu contributed equally to this work.

Yani Zhang, Yuzhou Yu, Huabao Chen are with the University of Science and Technology of China, Hefei 230027, China; and also with the Hefei Cancer Hospital of CAS, Institute of Health and Medical Technology, Hefei Institutes of Physical Science, Chinese Academy of Sciences, Hefei 230031, China.

Qiankun Li is with the College of Computing and Data Science (CCDS), Nanyang Technological University, 639798, Singapore.

Yuci Liang is with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen 518060, China

Junxin Chen is with the School of Software, Dalian University of Technology, Dalian 116024, China.

Jiayue Yin is with the Department of Prosthodontics, School of Stomatology, China Medical University, Shenyang 110002, China.

Jiayue Yin is the corresponding author (Email: jiayueyin.cmu@outlook.com).

I. Introduction

Tooth instance segmentation and identification in panoramic dental X-rays is a fundamental task in computer-aided dental diagnosis and treatment planning. Accurate delineation and numbering of individual teeth not only facilitate downstream applications such as caries localization, implant planning, and orthodontic assessment, but also enable structured reporting, clinical auditing, and longitudinal tracking of dental conditions [1], [2]. When coupled with the Fédération Dentaire Internationale (FDI) [3] numbering system, which encodes each tooth based on quadrant and position, segmentation results become interpretable and directly useful in clinical records. In clinical workflows, the FDI-coded segmentation maps further align with clinical workflows by facilitating dentist review, structured charting, and case archiving in electronic dental record systems.

Prior work spans both traditional rule-based pipelines and modern deep learning approaches. Early methods such as adaptive thresholding [4], [5], region growing [6], and marker-driven watershed [7] relied heavily on hand-crafted cues, including texture, morphology, and positional priors. Although these approaches achieved partial success, they were highly sensitive to image noise, low contrast, and anatomical variability, which substantially limited their robustness in clinical panoramic radiographs.

With the advent of deep learning, semantic segmentation has become the dominant paradigm [8]. Representative studies include coupled shape priors integrated with neural models [9], attention-based coarse-to-fine pipelines [10], patch- and ROI-based strategies employing U-Net [11]–[13], generative adversarial network (GAN) based boundary refinement [14], and dual subnetworks guided by distortion or boundary information [15]. More recent work has also proposed semi-supervised strategies, such as ellipse-based initialization, to alleviate annotation burdens [16]. Despite these advances, semantic segmentation inherently lacks the ability to separate instances and to assign tooth identities, which limits its clinical utility.

Instance segmentation frameworks address these shortcomings by providing tooth-level masks and supporting numbering to some extent. Early systems focused on detecting and segmenting individual teeth in panoramic images [2]. Exten-

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sions of Mask R-CNN have demonstrated strong performance when combined with large-scale pretraining and customized prediction heads [17], [18], with further adaptations for deciduous dentition [19] and heuristic post-hoc identification correction [20]. Two-stage pipelines, for example, DeepLab-v3 followed by FCN, have also been explored [21]. To improve data efficiency, semi- and self-supervised approaches have been investigated, including graph-based convolutional network (GCN) coarse-to-fine schemes [22] and self-supervised pretraining with SimMIM or UM-MAE [23]. Spatial priors have further been incorporated through center-based constraints and attention fusion mechanisms [24].

Nevertheless, most existing frameworks remain static, primarily optimized for permanent dentition, and often decouple numbering from segmentation. More importantly, they rarely consider bilateral symmetry or the hierarchical structure of dental arches, nor do they provide explicit mechanisms to handle anomalies such as supernumerary or missing teeth, or to verify output completeness. These limitations highlight the need for a system-level, symmetry-aware, and anatomically verified solution.

To address these limitations, we propose a reliable multi-stage system for tooth instance segmentation and numbering in panoramic radiographs. The system comprises three sequential stages: an entrance classifier to recognize supernumerary teeth and guide adaptive routing, a core segmentor for instance-level tooth segmentation and FDI-compliant numbering, and a missing tooth detector to identify and correct index gaps through post-hoc verification. Specifically, the supernumerary tooth classifier is implemented using MobileNetV3 [25], the missing tooth detector is based on YOLOv5 [26], and the core segmentation and numbering module is built upon a novel Symmetry-Aware Dual-branch Pyramid Network (SADP-Net). SADP-Net is specially designed as a dedicated instance segmentation network that tackles two key challenges: the morphological symmetry of dental arches and the anatomical scale variability between teeth. It introduces two novel components. First, a Symmetry-Aware Module (SymAM) captures bilateral structural consistency through feature mirroring and noise suppression. Second, a Dual-Branch Pyramid (DBP) architecture that integrates SymAM through either shallow or deep pathways, and conducts bidirectional multi-scale fusion to retain spatial details while enriching semantic representations.

Extensive experiments on three diverse datasets, including the Tufts Dental [27] and O^2PR [28] public benchmarks, as well as a collected multi-pathology dataset. Experimental results demonstrate superior segmentation and numbering performance compared to state-of-the-art baselines, while comprehensive ablation studies validate the contributions of the proposed components in improving robustness and anatomical consistency.

The main contributions of this work are summarized as follows:

- We propose a reliable multi-stage system for tooth instance segmentation and numbering in panoramic radiographs, incorporating adaptive routing, instance-level segmentation, and post-hoc correction for enhanced robustness and clinical applicability.
- We design a novel SADP-Net that improves boundary localization and symmetry-aware teeth segmentation with numbering through the proposed SymAM and DBP modules architecture.
- Our method achieves state-of-the-art results across three datasets and demonstrates the effectiveness of each proposed component in improving segmentation accuracy, numbering consistency, and robustness to anatomical variations.

The remainder of this paper is organized as follows. Section II reviews related studies in tooth segmentation and numbering. Section III details the architecture of the proposed multi-stage framework and its key components. Section IV describes the experimental protocols, datasets, and evaluation metrics. Section V presents the quantitative and qualitative results, along with an in-depth analysis of each component. Finally, Section VI concludes the paper and discusses potential directions for future work.

II. Related Work

A. Traditional-Method-Based Tooth Segmentation

Traditional methods primarily rely on thresholding, region-based, edge-based, clustering, watershed, and other approaches to extract features such as texture, morphological structure, and positional arrangement from dental panoramic X-ray images for tooth segmentation. Indraswari et al. [4] and Ahmad et al. [5] applied locally adaptive thresholding segmentation methods to tooth segmentation in dental panoramic X-ray images. This locally adaptive segmentation concept can be flexibly applied to different segmentation scenarios. Modi et al. [6] and Lurie et al. [29] utilized region-growing algorithms to identify tooth edge regions, then classified and grouped similar regions for tooth segmentation. Li et al. [30] combined level sets with support vector machines to improve tooth segmentation accuracy. Tuan [31] employed semi-supervised entropy-regularized fuzzy clustering with user-provided additional information to segment teeth. Datta and Chaki [7] used marker-driven watershed methods for tooth segmentation in dental panoramic X-ray images. However, traditional methods heavily rely on hand-crafted features and heuristic rules, and often exhibit limited robustness in complex clinical scenarios.

B. Tooth Semantic Segmentation

Semantic segmentation has become the predominant technique for analyzing panoramic dental radiographs. To address challenges such as blurred tooth boundaries and low contrast, Writz et al. [9] proposed a fully automated segmentation method that integrates coupled shape models with neural networks, incorporating gradient image features together with statistical priors on shape and spatial distribution. Zhao et al. [10] introduced a two-stage coarse-to-fine framework enhanced by attention mechanisms, where the approximate tooth regions are first localized, followed by refined segmentation of individual teeth. Similarly, Banar et al. [11] divided panoramic images into adjacent non-overlapping patches to extract regions of interest (ROIs), and subsequently applied a U-Net [12] to segment teeth within the ROIs. To achieve more accurate

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delineation of complex tooth boundaries, Cui et al. [14] proposed ToothPix, a GAN-based architecture that leverages wide residual blocks and an encoder–decoder generator, guided by a discriminator to jointly capture grayscale intensity and boundary features. Overlapping among adjacent teeth poses another challenge; Fariza et al. [13] addressed this issue by segmenting teeth in manually cropped ROI images using a U-Net. In addition, for handling severe misalignment of teeth, Zhang et al. [15] developed a dual-subnetwork architecture guided by feature distortion and boundary cues. Their approach employs Disout regularization in the encoder to generate distorted multi-scale feature maps, which are shared across subnetworks combining feature pyramid and U-Net structures for segmentation. Furthermore, Xie et al. [16] leveraged semi-supervised learning to predict the location and size of each tooth and generate initial ellipses from bounding boxes, alleviating the heavy manual annotation burden for large-scale datasets. Nevertheless, semantic segmentation remains limited by its inability to achieve precise spatial localization of individual teeth, thereby constraining its translational impact in clinical practice.

C. Tooth Instance Segmentation

Tooth instance segmentation has gradually become a research focus, as it enables not only the delineation of tooth regions but also their classification and numbering. Jader et al. [2] developed the first system for detecting and segmenting teeth in panoramic dental radiographs, incorporating instance-level information to achieve segmentation. To improve the segmentation accuracy of individual teeth, Lee et al. [17] performed hyperparameter tuning and network pretraining on the COCO dataset [32], and modified Mask R-CNN [18] to segment tooth images and assign classes to each tooth. Subsequently, Karaoglu et al. [20] proposed a heuristic algorithm to correct errors in tooth ID assignment by the network. Considering the segmentation of deciduous teeth, Pinheiro et al. [19] built an end-to-end deep learning framework based on Mask R-CNN for deciduous tooth segmentation and numbering.

Several two-stage tooth instance segmentation methods have also been proposed. For instance, Leite et al. [21] combined DeepLab-v3 with fully convolutional networks to detect and segment teeth through a two-stage neural network. Semi-supervised and self-supervised approaches have also been widely applied to tooth instance segmentation. To achieve more precise boundary delineation, Qiu et al. [22] introduced a semi-supervised framework that integrates a GCN with a coarse-to-fine strategy. Almalki et al. [23] leveraged self-supervised learning methods, including SimMIM and UM-MAE, to improve model efficiency and enhance understanding of limited dental radiographs. Considering the spatial priors inherent in the ordered arrangement of teeth, Li et al. [24] designed a center-based spatial prior-aware module to strengthen spatial constraints and employed an attention mechanism to effectively fuse spatial priors with long-range contextual features.

Different from existing methods, we exploit the intrinsic symmetrical structure of teeth by designing a Symmetry-Aware module, and introduce both shallow and deep pyramid fusion architectures. In addition, our method leverages the FDI international tooth numbering system, enabling tooth instance segmentation for both permanent and deciduous teeth.

III. Proposed Method

In this paper, we propose a reliable multi-stage system for tooth instance segmentation and numbering in panoramic radiographs, designed to ensure robustness and clinical applicability across diverse dental scenarios. As illustrated in Fig. 1, the proposed system integrates three main modules in a sequential pipeline: 1) anomaly-guided entrance routing via a supernumerary tooth classifier, 2) a Symmetry-Aware Dual-branch Pyramid Network (SADP-Net) for instance segmentation and numbering, and 3) a missing tooth detector for post-hoc correction. The following subsections detail each component.

A. Overall Framework

Given an input panoramic dental X-ray image $\boldsymbol{I} \in \mathbb{R}^{3\times H\times W}$, our proposed multi-stage system processes it through a sequential pipeline comprising classification, segmentation and numbering, and posterior correction. The framework can be formulated as:

$$\mathcal{S}: \boldsymbol{I} \xrightarrow{\mathcal{F}_{\text{cls}}} \hat{y}_{cls} \xrightarrow{\mathcal{F}_{\text{seg}}} (\boldsymbol{M}, \boldsymbol{N}) \xrightarrow{\mathcal{F}_{\text{corr}}} \{\boldsymbol{M}_{\text{final}}, \boldsymbol{N}_{\text{final}}\}. \quad (1)$$

First, a lightweight MobileNetV3 classifier [25], denoted as $\mathcal{F}_{\text{cls}}$, is employed to detect the presence of supernumerary teeth (i.e., multi-tooth anomaly), yielding a binary output $\hat{y}_{cls} \in \{0, 1\}$. Given the high reliability of this module, $\hat{y}_{cls}$ directly guides the routing strategy: images predicted to contain anomalies are forwarded to an anomaly-specific segmentation branch $\mathcal{P}_{\text{anom}}$, while normal cases are processed by the standard pipeline $\mathcal{P}_{\text{std}}$.

Subsequently, instance-level segmentation and tooth numbering are performed using our proposed SADP-Net, denoted as $\mathcal{F}_{\text{seg}}$. This stage produces an instance mask $\boldsymbol{M} \in \{0, 1\}^{H\times W\times K}$ and a corresponding per-pixel numbering map $\boldsymbol{N} \in \{1, \ldots, 32\}^{H\times W}$.

To improve the robustness of the output, a YOLOv5-based [26] missing tooth detector $\mathcal{F}_{\text{miss}}$ is utilized in the posterior correction stage. Rather than serving as a primary decision module, $\mathcal{F}_{\text{miss}}$ acts as an auxiliary verifier by predicting a tooth presence vector $\boldsymbol{M}_{\text{miss}} \in \{0, 1\}^{32}$. This vector helps identify potential segmentation inconsistencies (e.g., undetected or misnumbered teeth), and guides a lightweight correction module $\mathcal{F}_{\text{corr}}$ to refine the output.

The overall system thus integrates reliable anomaly classification with specialized segmentation and auxiliary correction, ensuring both robustness and flexibility for clinical panoramic dental analysis.

B. Supernumerary Teeth Classification

To identify supernumerary teeth (hyperdontia) at the system input stage, we formulate a binary classification module that processes panoramic dental X-ray images. This module provides a routing signal for downstream processing.

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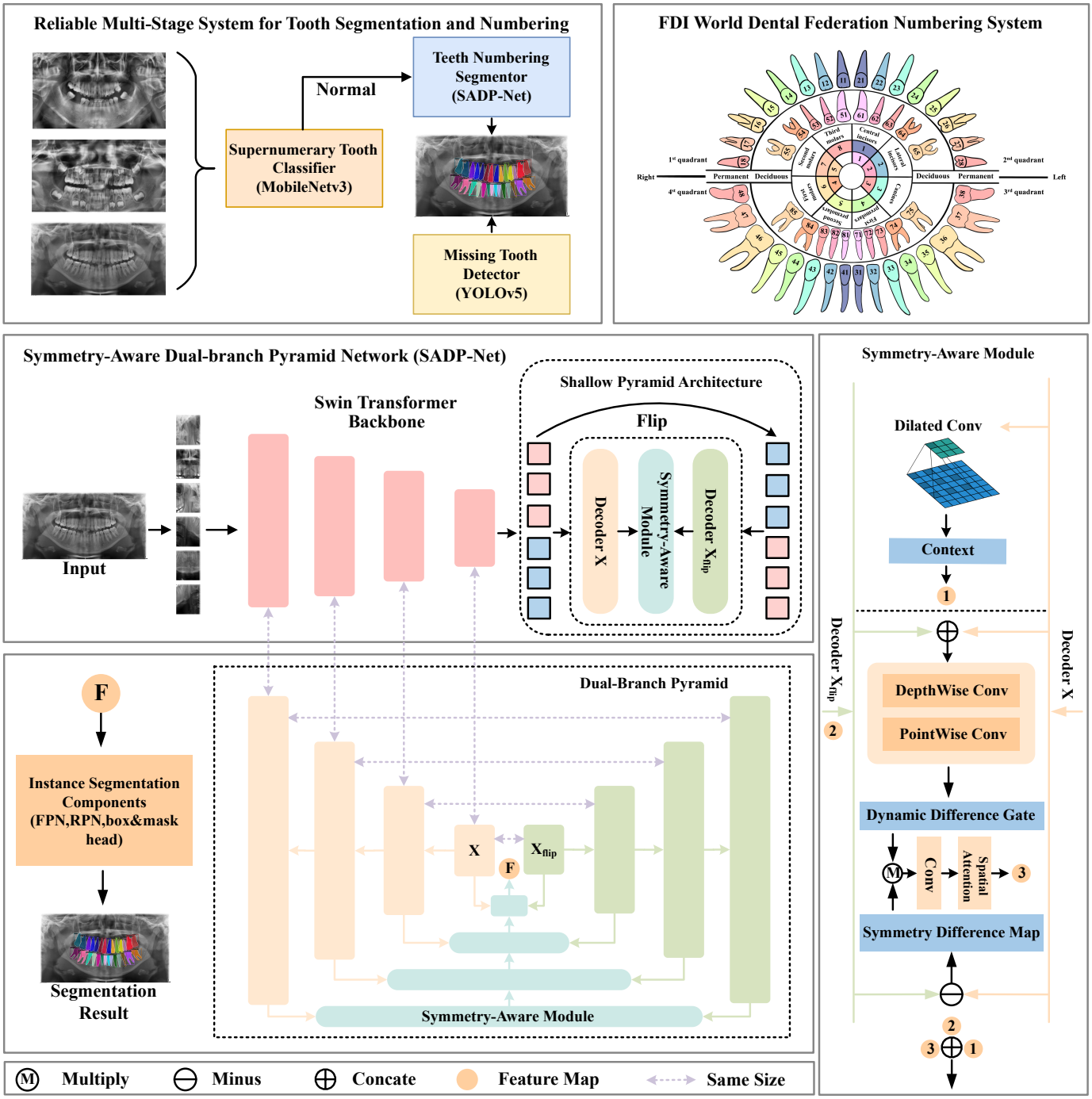

Fig. 1: Overview of the proposed reliable multi-stage tooth instance segmentation and numbering system. It consists of a supernumerary tooth classifier, the SADP-Net for instance segmentation and FDI numbering, and a missing tooth detector for correction. SADP-Net introduces a SymAM and a DBP for symmetry-enhanced, multi-scale feature fusion.

1) Network Architecture: We adopt MobileNetV3-Small [25] as the backbone $\mathcal{F}_{\text{anom}}$ due to its high efficiency and competitive performance. The input image $\boldsymbol{I} \in \mathbb{R}^{3\times H\times W}$ is first encoded by a series of MobileNetV3 blocks:

$$\boldsymbol{F}_0 = \text{Conv}_{3\to 16}^{3\times 3}(\boldsymbol{I}), \tag{2}$$

$$\boldsymbol{F}_k = \mathcal{B}_k(\boldsymbol{F}_{k-1}), \quad \forall k \in \{1,\dots,5\}, \tag{3}$$

where each $\mathcal{B}_k$ denotes a MobileNetV3 block composed of a depthwise separable convolution, batch normalization, Hard-Swish activation, and a SE module [33]:

$$\mathcal{B}_k(\mathbf{X}) = \text{HSwish}\left(\text{BN}\left(\text{DWConv}_{d_k}^{3\times 3}(\mathbf{X})\right)\right) \otimes \text{SE}(\mathbf{X}). \tag{4}$$

2) Dynamic Classification Head: The final feature $\boldsymbol{F}_5$ is globally pooled and passed through a linear projection and activation layer:

$$\phi(\boldsymbol{x}) = \text{HSwish}\left(\text{BN}\left(\mathbf{W}_{\text{proj}}\boldsymbol{x}\right)\right), \quad \mathbf{W}_{\text{proj}} \in \mathbb{R}^{1280\times 256}, \tag{5}$$

$$\hat{y}_{cls} = \text{sign}\left(\mathbf{w}^T\phi(\text{GAP}(\boldsymbol{F}_5)) + b - \tau\right), \tag{6}$$

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where $\tau = 0.7$ is a clinically derived decision threshold. If $\hat{y}_{cls} = 1$, the image is routed to the anomaly-specific branch.

3) Training and Optimization: We employ binary cross-entropy as the loss function:

$$\mathcal{L}_{\text{anom}} = -\left[y \log(p_{\text{anom}}) + (1-y)\log(1-p_{\text{anom}})\right], \quad (7)$$

where $y \in \{0,1\}$ is the ground truth label and p_{anom} is the predicted probability before thresholding. To improve generalization, standard data augmentation techniques such as random horizontal flipping, rotation, and contrast adjustment are applied during training.

C. Symmetry-Aware SADP-Net for Segmentation and Numbering

The core segmentation and numbering module, denoted as $\mathcal{F}_{\text{seg}}$, is implemented using our proposed SADP-Net, a specialized instance segmentation network tailored for panoramic dental radiographs. SADP-Net is built upon the Mask R-CNN framework [18], with a Swin Transformer [34] backbone for multi-scale feature extraction, and it further incorporates two key components: a symmetry-aware feature enhancement module and a dual-branch pyramid fusion architecture.

1) Symmetry-Aware Feature Enhancement: Given an intermediate feature map $\boldsymbol{F}_{\text{enc}} \in \mathbb{R}^{C\times H\times W}$ from the encoder backbone, the symmetry-aware module captures bilateral symmetry through horizontal flipping and difference modeling. Specifically, we construct a flipped version $\boldsymbol{F}_{\text{flip}}$ as:

$$\boldsymbol{F}_{\text{flip}}[c,h,w] = \boldsymbol{F}_{\text{enc}}[c,h,W-w+1]. \quad (8)$$

To localize tooth boundaries and suppress asymmetric noise, we introduce a dynamic gating mechanism:

$$\boldsymbol{G} = \sigma\left(\text{Conv}^{3\times3}(\text{concat}(\boldsymbol{F}_{\text{enc}}, \boldsymbol{F}_{\text{flip}}))\right), \quad (9)$$

$$\boldsymbol{D} = \boldsymbol{G} \odot (\boldsymbol{F}_{\text{enc}} - \boldsymbol{F}_{\text{flip}}), \quad (10)$$

where $\boldsymbol{D}$ represents the gated symmetry difference map.

We refine this difference through edge-aware enhancement using depthwise separable convolution and spatial attention:

$$\boldsymbol{E} = \text{ReLU}\left(\text{BN}\left(\text{Conv}^{1\times1}\left(\text{DWConv}^{3\times3}(\boldsymbol{D})\right)\right)\right), \quad (11)$$

$$\boldsymbol{M}_{\text{att}} = \sigma\left(\text{Conv}^{7\times7}\left(\text{AvgPool}_c(\boldsymbol{E}) + \text{MaxPool}_c(\boldsymbol{E})\right)\right), \quad (12)$$

$$\boldsymbol{E}_{\text{att}} = \boldsymbol{E} \otimes \boldsymbol{M}_{\text{att}}. \quad (13)$$

Meanwhile, we apply dilated convolution on $\boldsymbol{F}_{\text{enc}}$ to extract contextual features:

$$\boldsymbol{F}_{\text{ctx}} = \text{ReLU}\left(\text{BN}\left(\text{DilatedConv}^{3\times3}_{r=2}(\boldsymbol{F}_{\text{enc}})\right)\right). \quad (14)$$

The resulting contextual feature $\boldsymbol{F}_{\text{ctx}}$ is then concatenated with the flipped and edge-aware features to form a joint representation, which is refined by a convolutional block followed by normalization and activation to produce the final symmetry-aware feature:

$$\boldsymbol{F}_{\text{sym}} = \text{ReLU}\left(\boldsymbol{F}_{\text{enc}} + \text{BN}\left(\text{Conv}^{3\times3}(\text{concat}(\boldsymbol{F}_{\text{ctx}}, \boldsymbol{F}_{\text{flip}}, \boldsymbol{E}_{\text{att}}))\right)\right). \quad (15)$$

2) Dual-Branch Pyramid Fusion: To capture features across multiple anatomical scales, we construct a dual-branch top-down pyramid from the output of the backbone's final stage. Let $\boldsymbol{F}_4^{(o)}$ and $\boldsymbol{F}_4^{(f)}$ denote the original and flipped branch features at the coarsest resolution. Both branches are progressively upsampled:

$$\boldsymbol{F}_i^{(o)} = \text{UpConv}^{4\times4}(\boldsymbol{F}_{i+1}^{(o)}), \quad (16)$$

$$\boldsymbol{F}_i^{(f)} = \text{Flip}\left(\text{UpConv}^{4\times4}(\boldsymbol{F}_{i+1}^{(f)})\right), \quad (17)$$

for $i \in \{3,2,1\}$. At each level i, the symmetry-aware module fuses both branches:

$$\boldsymbol{P}_i = \mathcal{F}_{\text{sym}}(\boldsymbol{F}_i^{(o)}, \boldsymbol{F}_i^{(f)}). \quad (18)$$

To complete the feature hierarchy, we adopt a top-down fusion strategy that recursively integrates features:

$$\boldsymbol{H}_i = \text{ReLU}\left(\text{BN}\left(\text{Conv}^{3\times3}(\text{concat}(\boldsymbol{P}_i, \text{Up}(\boldsymbol{H}_{i+1})))\right)\right). \quad (19)$$

The final feature map $\boldsymbol{H}_1$ encodes multi-scale, symmetry-aware representations for tooth instances.

3) Prediction and Loss: The final feature map $\boldsymbol{H}_1$ is shared by two decoder heads: a segmentation head that predicts instance masks $\boldsymbol{M}_{\text{seg}} \in \{0,1\}^{H\times W\times K}$ and a numbering head that assigns tooth indices $\boldsymbol{N}_{\text{label}} \in \{1,\ldots,32\}^{H\times W}$. The network is trained with two pixel-wise cross-entropy losses:

$$\mathcal{L}_{\text{seg}} = -\sum_{i,j} \log P_{\text{seg}}(i,j), \quad (20)$$

$$\mathcal{L}_{\text{num}} = -\sum_{i,j} \log P_{\text{num}}(i,j), \quad (21)$$

where $P_{\text{seg}}(i,j)$ and $P_{\text{num}}(i,j)$ are the predicted class probabilities at pixel (i,j).

D. Missing Tooth Detection and Correction

To enhance anatomical consistency and address potential segmentation omissions, we incorporate an auxiliary missing tooth detector $\mathcal{F}_{\text{miss}}$ and a correction module $\mathcal{F}_{\text{corr}}$ as a lightweight postprocessing step.

1) Tooth Presence Estimation: We adapt the YOLOv5 [26] architecture to estimate tooth presence across 32 indices. Given the fused feature map from $\mathcal{F}_{\text{seg}}$, the detector predicts a binary presence vector $\boldsymbol{M}_{\text{miss}} \in \{0,1\}^{32}$, where each element is computed as:

$$p_t = \sigma\left(\boldsymbol{w}_t^T \boldsymbol{f}^{(t)} + b_t\right), \quad \boldsymbol{M}_{\text{miss}}[t] = \mathbb{1}_{p_t > \tau_m}, \quad (22)$$

for tooth index $t \in \{1,\ldots,32\}$, where $\boldsymbol{f}^{(t)}$ denotes the region-level feature, σ is the sigmoid function, and τ_m is the confidence threshold.

2) Correction Strategy: The presence vector $\boldsymbol{M}_{\text{miss}}$ is used to verify the completeness of the predicted numbering map $\boldsymbol{N}_{\text{label}}$. For any missing index (i.e., $M_{\text{miss}}[t] = 0$), a correction is triggered:

$$\boldsymbol{N}_{\text{final}} = \mathcal{F}_{\text{corr}}(\boldsymbol{N}_{\text{label}}, \boldsymbol{M}_{\text{miss}}), \quad (23)$$

which realigns the numbering sequence and optionally refines local boundaries using surrounding topology. The final corrected segmentation output is denoted as $\boldsymbol{M}_{\text{final}}$.

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TABLE I: Summary of datasets used in this study.

Dataset	Resolution	Total Images	Split	Special Use
Tufts Dental [27]	1615×840	1,000	70/10/20	Tooth segmentation
O^2PR [28]	2440×1292	1,996	85/15	Tooth segmentation
Oral Multi-pathology [35]	Var-Res	1,000	70/10/20	129 hyperdontia / 129 normal (classification), 300 hypodontia (detection)

As an auxiliary component, this correction mechanism operates only when low-confidence regions are detected, ensuring minimal intervention to the primary SADP-Net predictions.

IV. Experiment Configuration

A. Datasets

We conduct experiments on three panoramic dental X-ray datasets, each capturing different aspects of dental structure and anomaly presentation. Table I summarizes the key properties of the datasets used in this study.

1) Tufts Dental Database [27]: is a publicly available benchmark containing 1,000 panoramic dental X-ray images with expert annotations for both permanent and deciduous teeth, covering normal and abnormal conditions. Each image has a resolution of 1615×840 pixels. We randomly divide the dataset into 70% for training, 10% for validation, and 20% for testing.

2) O^2PR Dataset [28]: is constructed using a human-in-the-loop strategy and provides complete tooth segmentation annotations following the FDI World Dental Federation notation. The dataset comprises 1,996 images with a resolution of 2440×1292 pixels. Of these, 1,696 images are used for training and 300 for validation. Since the distributed version does not include a dedicated test set, the validation set is also used for evaluation.

3) Oral Multi-pathology Dataset [35]: is a clinically collected dataset obtained in collaboration with Pujiang Dental Hospital. It comprises 1,000 high-resolution panoramic dental X-ray images. Among them, 129 images contain supernumerary teeth (hyperdontia), and another 129 images contain normal dentition without abnormalities. These 258 images are used for training and evaluating our supernumerary teeth classification module. In addition, 300 images with missing teeth (hypodontia) are used to train and evaluate our auxiliary missing tooth detection and correction module. Representative panoramic samples from the dataset, including normal, supernumerary, and missing-tooth cases, are shown in Fig. 2. All data collection procedures were approved by the Medical Ethics Committee of Pujiang Dental Hospital (Approval No. 2022-0102) and conducted in accordance with relevant ethical guidelines.

B. Evaluation Metrics

We adopt task-specific evaluation metrics tailored to the functional objectives of each module. For supernumerary tooth classification, we report accuracy, precision, recall, and F1-score to assess binary classification performance at the image level. For segmentation and numbering, model performance is evaluated using standard MS COCO metrics, including mean Average Precision (mAP) as well as AP at IoU thresholds of 0.50 (AP50) and 0.75 (AP75), which jointly reflect detection precision and localization quality. Additionally, accuracy, recall, and F1-score are incorporated to further assess model performance. For missing tooth detection, we follow the COCO detection protocol, computing AP50, precision, and recall based on the IoU between predicted bounding boxes and ground truth annotations. All metrics are consistently reported on the respective validation sets to ensure reliable and fair comparison across modules.

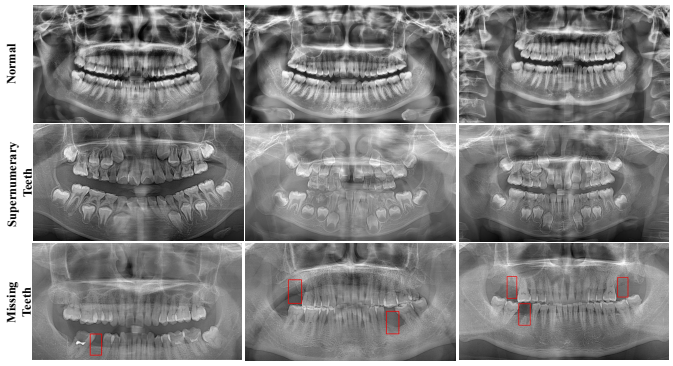

Fig. 2: Representative panoramic radiographs from the Oral Multi-pathology dataset, including normal, supernumerary, and missing-tooth cases. Red boxes indicate annotated missing-tooth regions.

1) Per-Tooth Confusion Matrix Analysis: To further examine the per-tooth segmentation and numbering performance, Fig. 4 presents the normalized confusion matrix of the proposed *Ours-D* model on the O^2PR dataset. Each row corresponds to a ground-truth tooth index following the FDI convention, and each column denotes the predicted label.

C. Implementation Details

All experiments were carried out using the PyTorch deep learning framework with the mmdetection toolbox [36]. Model training and inference were conducted on a single NVIDIA RTX 4090 GPU. Unless otherwise specified, each model was optimized using AdamW with an initial learning rate of 1×10^{-4}, which was reduced by a factor of 10 at the 24th and 40th epochs. Training lasted for a total of 50 epochs with a batch size of 4. In addition, during the testing phase, the bbox score threshold and NMS threshold of RCNN were set to 0.05 and 0.5, respectively.

To standardize input dimensions across datasets while minimizing geometric distortion, all panoramic radiographs were resized to 1024×512 resolution. Padding with black pixels was used to maintain the original aspect ratio during resizing. For different sub-networks within the multi-stage framework,

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the input configurations were adjusted according to their functional requirements: the supernumerary tooth classification module processes 224×224 images using MobileNetV3-Small; the SADP-Net segmentation and numbering module uses 1024×512 panoramic inputs; and the missing-tooth detection module is trained with 640×640 inputs based on the YOLOv5m architecture. The Tufts database provided annotations in a COCO-inspired JSON format, which required correction before training. To resolve issues such as fragmented masks, single-pixel bounding boxes, and edge artifacts, we applied morphological closing followed by Gaussian smoothing. Refined bounding boxes were then extracted from cleaned masks to improve quality.

V. Results and Discussions

In this section, we present comprehensive quantitative and qualitative evaluations of the proposed framework across three tasks: tooth classification, instance segmentation and numbering, and missing tooth detection. We first analyze the segmentation and numbering performance with quantitative and qualitative results, then conduct ablation studies to validate each module's contribution.

TABLE II: Quantitative results for supernumerary tooth classification.

Model	F1	Acc	Precision	Recall	Params
ResNet-18 [37]	94.37	94.34	94.47	94.34	11.7M
MobileNetV2 [38]	94.42	94.34	95.11	94.34	3.4M
MobileNetV3 [25]	**100.00**	**100.00**	**100.00**	**100.00**	**2.5M**
EfficientNet-B0 [39]	**100.00**	**100.00**	**100.00**	**100.00**	5.3M

A. Supernumerary Tooth Classification Results

We evaluate five representative classification backbones on the task of identifying supernumerary teeth, with results summarized in Table II. Each model is assessed based on F1-score, accuracy, precision, and recall, with the number of parameters also listed for reference.

Both MobileNetV3 [25] and EfficientNet-B0 [39] achieve perfect classification performance across all evaluation metrics (100.00%), highlighting their strong discriminative capacity even on a relatively small dataset. Notably, MobileNetV3 attains these results with only 2.5M parameters, making it more lightweight and efficient than EfficientNet-B0.

MobileNetV2 [38] and ResNet-18 [37] also show competitive performance, both exceeding 94% in all metrics while remaining computationally efficient.

Given the simplicity of the classification task and the need for efficient processing in a real-time diagnostic pipeline, we adopt MobileNetV3 [25] as the backbone for the supernumerary tooth classification module. Its compact size and excellent performance make it particularly well-suited for serving as a front-end filtering mechanism in our integrated tooth segmentation and numbering system.

B. Segmentation and Numbering Results

The proposed method is evaluated and compared with 11 state-of-the-art instance segmentation approaches on two panoramic dental X-ray image datasets, including IS Panoramic [2], ToothSNN [40], Cascade R-CNN [41], SOLOv2 [42], QueryInst [43], HTC [44], DetectoRS [45], Mask2former [46], DentalMIM [23], Mask R-CNN [18], and SPGTNet [24]. Among these, results on the Tufts Dental Database include only permanent teeth, while results on the O^2PR Dataset encompass both permanent and deciduous teeth. Our proposed shallow pyramid architecture *Ours-S* employs solely the Symmetry-Aware Module, whereas the deep pyramid architecture *Ours-D* integrates both the Symmetry-Aware Module and the Dual-Branch Pyramid.

1) Tufts Dental Database Result: As listed in Table III, the previous state-of-the-art method is SPGTNet [24], which integrates the HTC [44] segmentation head, achieving mAP, AP50, and AP75 scores of 62.2%, 97.3%, and 76.1% respectively on the Tufts Dental Database. Our proposed *Ours-D* method attains 66.2% mAP, 97.3% AP50, and 85.3% AP75. Compared to SPGTNet [24], *Ours-D* achieves a 4.0% improvement in mAP and a 9.2% gain in AP75. Furthermore, our method significantly outperforms other CNN-based and Transformer-based approaches, with this superior instance segmentation performance attributed to the strategic integration of the Symmetry-Aware Module and Dual-Branch Pyramid. By incorporating the inherent symmetrical structure of teeth in panoramic X-ray images, capturing multi-scale feature representations of dental targets, and leveraging Swin Transformer's four-stage hierarchical architecture and its powerful representational capacity, our approach substantially enhances tooth instance segmentation capabilities. Notably, *Ours-D* achieves optimal performance across both AP50 and AP75 metrics, highlighting its robustness in precisely detecting tooth instances under varying IoU thresholds. Simultaneously, *Ours-D* also achieves the highest scores in Accuracy, Recall, and F1-score metrics.

Compared to the *Ours-S* method, *Ours-D* demonstrates further improvements: +1.0% in mAP, +0.3% in AP50, and +1.6% in AP75. This confirms the exceptional capability of the Dual-Branch Pyramid in effectively fusing spatial details from shallow features and semantic information from deep features, thereby substantially enhancing tooth instance segmentation performance. As shown in Fig. 3(a), the proposed *SADP-Net* attains the highest performance across all core metrics (mAP, AP50, AP75, and F1-score) on the Tufts Dental Database [27], exhibiting consistently superior results compared with competing methods.

2) O^2PR Dataset Result: As listed in Table IV, experimental results on the O^2PR Dataset validate the superiority of our proposed method for tooth instance segmentation involving both permanent and deciduous teeth. Our method approaches outperform all comparative methods across accuracy metrics including mAP, AP50, AP75 demonstrating their capability to generate more precise and refined segmentation masks. This conclusively highlights the effectiveness of integrating the inherent symmetrical structure of teeth in panoramic X-

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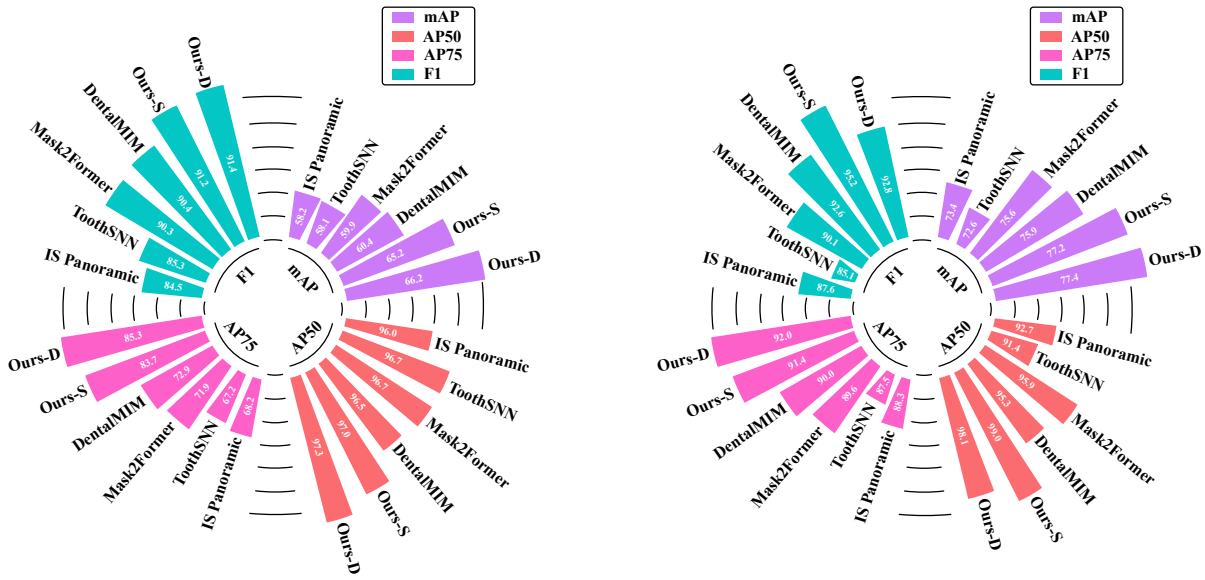
TABLE III: Quantitative comparison with state-of-the-art methods on the Tufts dental database.

Method	Backbone	Params (M)	mAP	AP50	AP75	Accuracy	Recall	F1
IS Panoramic [2]	ResNet	62.9	58.2	96.0	68.2	99.1	84.1	84.5
ToothSNN [40]	ResNet	66.5	58.1	96.7	67.2	99.2	84.6	85.3
Cascade R-CNN [41]	ResNet	95.9	59.3	95.9	71.3	99.4	88.3	89.6
SOLOv2 [42]	ResNet	65.6	57.0	91.8	68.9	93.9	-	-
QueryInst [43]	ResNet	191.3	58.5	95.7	69.0	99.4	87.4	89.2
HTC [44]	ResNet	140.0	61.5	97.3	74.4	99.5	89.9	90.9
DetectoRS [45]	ResNet	196.7	58.8	97.2	68.7	99.5	89.1	90.1
Mask2Former [46]	Swin	68.7	59.9	96.7	71.9	99.5	89.5	90.3
DentalMIM [23]	Swin	106.2	60.4	96.5	72.9	99.5	89.8	90.4
Mask R-CNN [18]	ViT	43.5	57.6	95.5	68.0	98.9	80.4	81.1
SPGTNet [24]	ViT	53.9	60.7	96.7	73.1	99.4	89.0	89.4
SPGTNet-HTC [24]	BEiT	171.9	62.2	97.3	76.1	99.5	89.8	90.5
Ours-S	Swin	80.1	65.2	97.0	83.7	99.5	90.6	91.2
Ours-D	Swin	116.7	**66.2**	**97.3**	**85.3**	**99.5**	**90.9**	**91.4**



(a) Tufts Dental Database Experiment Result **(b) O^{2}PR Dataset Experiment Result**

Fig. 3: Multi-metric performance comparison (mAP, AP50, AP75, F1-score). (a) Tufts Dental Database Experiment Result. (b) O^{2}PR Dataset Experiment Result.

TABLE IV: Quantitative comparison with state-of-the-art methods on the O^{2}PR dataset.

Method	Backbone	Params (M)	mAP	AP50	AP75	Accuracy	Recall	F1
IS Panoramic [2]	ResNet	62.9	73.4	92.7	88.3	99.7	86.6	87.6
ToothSNN [40]	ResNet	66.5	72.6	91.4	87.5	99.7	83.7	85.1
Cascade R-CNN [41]	ResNet	95.9	73.5	92.3	86.4	99.8	87.4	89.0
SOLOv2 [42]	ResNet	65.6	69.8	87.3	81.5	96.1	-	-
QueryInst [43]	ResNet	191.3	75.7	95.6	90.2	99.8	89.6	90.8
HTC [44]	ResNet	140.0	76.5	96.4	90.5	99.9	93.3	94.4
DetectoRS [45]	ResNet	196.7	75.6	95.5	89.5	99.9	91.7	92.6
Mask2Former [46]	Swin	68.7	75.6	95.9	89.6	99.8	90.4	90.1
DentalMIM [23]	Swin	106.2	75.9	95.3	90.0	99.9	92.7	92.6
Mask R-CNN [18]	ViT	43.5	73.2	92.8	86.1	99.7	82.5	83.2
Ours-S	Swin	80.1	77.2	**99.0**	91.4	99.9	**94.7**	**95.2**
Ours-D	Swin	116.7	**77.4**	98.1	**92.0**	**99.9**	91.7	92.8

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ray images and capturing multi-scale feature representations for enhancing tooth instance segmentation.

Specifically, our method achieves an impressive AP50 of 99.0%, surpassing the previous state-of-the-art HTC [44] method by 2.6%, while also improving AP75 by 1.5%, further demonstrating the high-precision capability of our approach in detecting tooth instances under varying IoU thresholds. Additionally, our method attains significantly higher Recall and F1-score compared to other approaches, indicating superior performance in accurately identifying tooth instances. As shown in Fig. 3 (b), our method achieves leading performance across all four core metrics, with the most notable improvements observed in AP50 and F1-score.

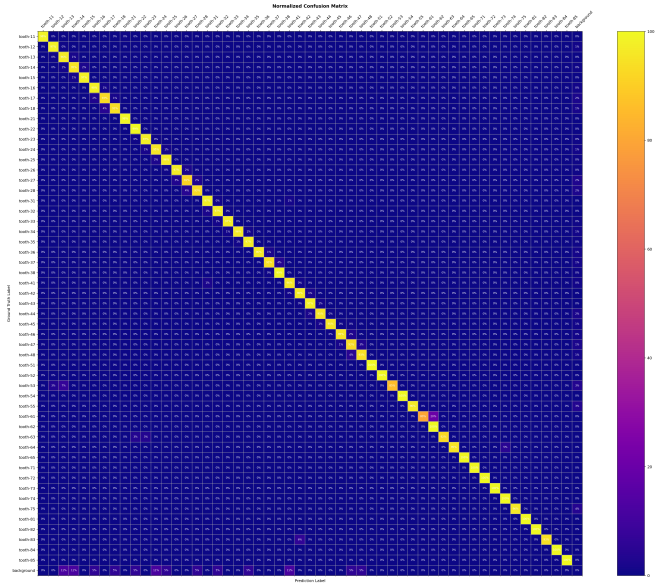

Fig. 4: Normalized per-tooth confusion matrix of the proposed *Ours-D* model on the O^{2}PR dataset (IoU threshold = 0.5). The near-diagonal structure indicates high per-tooth accuracy.

3) Per-Tooth Confusion Matrix Analysis: To further examine the per-tooth segmentation and numbering performance, Fig. 4 presents the normalized confusion matrix of the proposed *Ours-D* model on the O^{2}PR dataset. Each row corresponds to a ground-truth tooth index following the FDI convention, and each column denotes the predicted label.

As shown, most diagonal elements are close to 100%, indicating highly consistent tooth-level segmentation and numbering across the full dental arch. Occasional off-diagonal entries appear mainly in the deciduous teeth region, reflecting minor index ambiguities between symmetrically positioned teeth or small low-contrast deciduous structures. These observations are consistent with our prior analysis that anatomical similarity and reduced pixel occupancy can lead to confusion in low-contrast areas.

Overall, the confusion matrix confirms that *Ours-D* achieves strong per-tooth accuracy and robust numbering consistency.

C. Qualitative Results

Fig. 5 shows qualitative comparisons on the O^{2}PR dataset. We compare our method with top CNN-based (QueryInst [43], HTC [44]) and Transformer-based (Mask2Former [46], DentalMIM [23]) approaches.

In low-contrast regions with deciduous teeth (Col. 1), QueryInst confuses adjacent teeth, DentalMIM misassigns tooth numbers, HTC exhibits positional drift, and Mask2Former misses detections. Our method accurately localizes and numbers teeth. For overlapping teeth (Col. 2), CNN methods fail to detect obscured teeth, while Transformer methods produce numbering errors; our approach correctly segments and numbers all teeth. In implant cases (Col. 3), prior methods misclassify guide drills as teeth and misassign numbers; our method distinguishes implants from real teeth and maintains correct numbering. For root fragments (Col. 4), previous methods detect only partial fragments with numbering errors, whereas our method detects all fragments with precise labeling.

Overall, superior performance arises from the Symmetry-Aware Module, which resolves deciduous tooth mismatches and symmetric occlusion ambiguities, and the Dual-Branch Pyramid, which enhances detection of small teeth, overlapping teeth, and root fragments through multi-scale feature fusion. Their synergy also improves robustness against adjacent tooth misassignment in the presence of foreign objects.

D. Missing Tooth Detection Results

We evaluate the performance of the missing tooth detection task, which serves as a post-hoc correction stage in the overall multi-stage system. Experiments are conducted using YOLOv5m on the Oral Multi-Pathology Dataset, with quantitative results listed in Table V. The model achieves an AP50 of 60.4%, a precision of 66.7%, and a recall of 56.4%, indicating that the detector provides reliable performance in identifying missing teeth.

TABLE V: Quantitative results for missing tooth detection

Method	**Oral Multi-pathology Dataset**			
	AP50	Precision	Recall	Params
YOLOv5m [26]	60.4	66.7	56.4	21.2M

Qualitative results are shown in Fig. 6, where the predicted outputs show high visual consistency with the ground truth annotations. The detector accurately localizes and identifies missing teeth across diverse cases, confirming its effectiveness in providing reliable correction signals to improve the overall robustness of the integrated diagnostic system.

E. Ablation Study

1) Effect of Symmetry-Aware Module and Dual-Branch Pyramid: To investigate the individual and combined contributions of the Symmetry-Aware Module and the Dual-Branch Pyramid, we conducted ablation studies based on the Mask R-CNN framework [18] with a Swin Transformer backbone. Specifically, we denote the integration of the Dual-Branch Pyramid as *DF* and the inclusion of the horizontal flipping-based symmetry modeling as *Flip*. The baseline model contains neither component. We define the variant with only *DF* as

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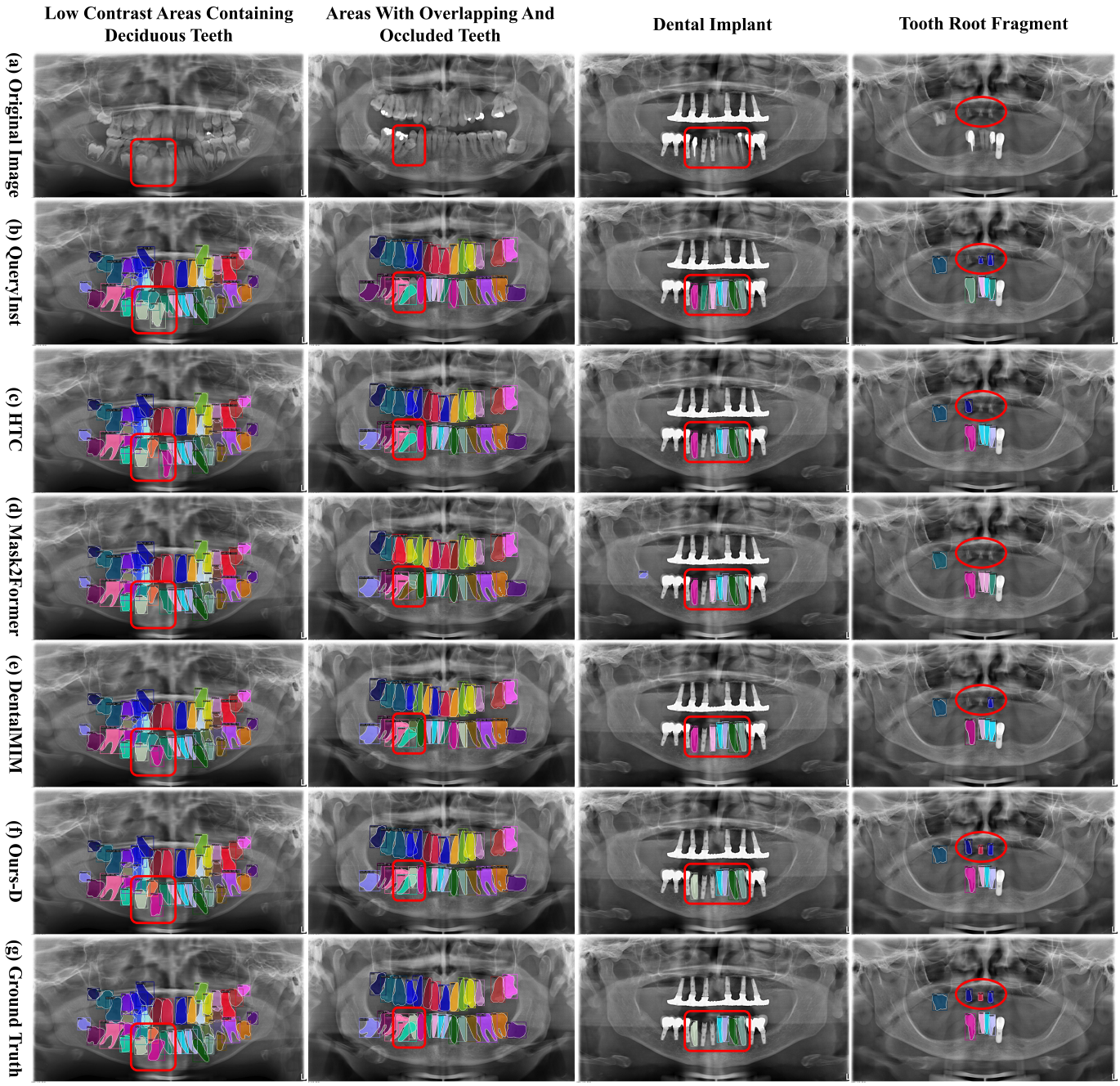

Fig. 5: Qualitative results compared with the state-of-the-art tooth instance segmentation methods. (a) Original Image. (b) QueryInst. (c) HTC. (d) Mask2Former. (e) DentalMIM. (f) Ours-D. (g) Ground Truth.

Dual-Branch FPN, and the version with only *Flip* as *SADP-Net*$_{Shallow}$. The full model with both modules is denoted as *SADP-Net*$_{Deep}$.

Table VI summarizes the performance of each configuration on the Tufts Dental Database. Integrating DF alone improves mAP, AP50, and AP75 by 0.7%, 0.9%, and 1.5%, respectively, highlighting the module's capacity to enhance multi-scale fusion for dense dental instances. Incorporating Flip achieves more substantial gains, improving mAP by 1.2% and AP75 by 5.0%, validating the effectiveness of leveraging bilateral symmetry in dental anatomy. The complete model SADP-Net$_{\text{Deep}}$ further improves mAP and AP50 over the shallow variant, demonstrating the complementary nature of symmetry-aware modeling and dual-branch fusion.

TABLE VI: Ablation study of symmetry modeling (Flip) and dual-branch fusion (DF).

Method	Flip	DF	mAP	AP50	AP75
Baseline			58.1	89.0	70.5
Dual-Branch FPN		✓	58.8	89.9	72.0
SADP-Net$_{\text{Shallow}}$	✓		59.3	90.7	**75.5**
SADP-Net$_{\text{Deep}}$	✓	✓	**59.5**	**91.8**	73.6

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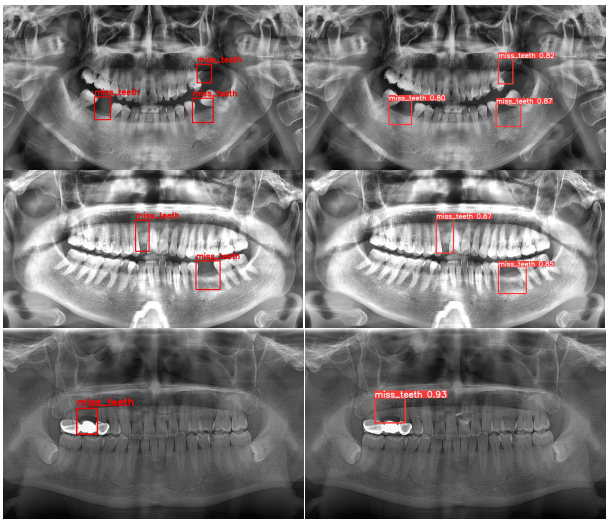

(a) Ground Truth **(b) Prediction**

Fig. 6: Visual performance of missing tooth detection on the Oral Multi-Pathology Dataset. (a) Ground Truth. (b) Predicted.

2) Effect of Symmetry and Edge Features: To assess the effectiveness of different feature cues in modeling dental symmetry, we evaluate the contributions of symmetry-aware features and edge-aware features within the proposed Symmetry-Aware Module. Specifically, we consider three configurations based on a Mask R-CNN [18] baseline with a Swin Transformer backbone: introducing only edge features (*+Edge Feature*), introducing only flip-based symmetry features (*+Flip Feature*), and integrating both (*All Components*), which constitutes the complete Symmetry-Aware Module.

As shown in Table VII, both edge and flip features individually enhance segmentation performance compared to the baseline. Notably, flip-based symmetry features yield a more significant improvement in AP75, reflecting their ability to capture fine-grained spatial correspondence between symmetrical dental structures. The full configuration (*All Components*) achieves the best results across all metrics, with a 1.2% increase in mAP, a 1.7% gain in AP50, and a substantial 5.0% improvement in AP75 over the baseline. These results confirm the complementary roles of edge and symmetry features and demonstrate the effectiveness of the proposed dual-branch strategy in capturing anatomical symmetry for dental instance segmentation.

TABLE VII: Ablation study of symmetry features and edge features in the symmetry-aware module.

Method	Edge Feature	Flip Feature	mAP	AP50	AP75
Baseline			58.1	89.0	70.5
+Edge Feature	✓		58.2	89.2	70.6
+Flip Feature		✓	58.7	89.3	74.0
All Components	✓	✓	**59.3**	**90.7**	**75.5**

3) Effect of Difference Gate and Spatial Attention: The Symmetry-Aware Module incorporates two key components for enhanced edge feature extraction: a dynamic difference gate and a spatial attention mechanism. To assess their individual contributions, we conducted comparative ablation experiments, and the results are summarized in Tables VIII and IX.

TABLE VIII: Ablation study of the dynamic difference gate in edge feature extraction.

Method	mAP	AP50	AP75
Baseline	58.8	90.3	72.6
+ Difference Gate	**59.3**	**90.7**	**75.5**

TABLE IX: Ablation study of the spatial attention mechanism in edge feature extraction.

Method	mAP	AP50	AP75
Baseline	58.4	90.1	72.9
+ Spatial Attention	**59.3**	**90.7**	**75.5**

As listed in Table VIII, introducing the dynamic difference gate into the baseline model leads to consistent improvements across all metrics, with gains of 0.5% in mAP, 0.4% in AP50, and 2.9% in AP75. These results confirm the gate's effectiveness in preserving meaningful edge responses while suppressing background noise and irrelevant structures.

Table IX reports the impact of incorporating spatial attention. Compared to the baseline, this mechanism improves mAP, AP50, and AP75 by 0.9%, 0.6%, and 2.6%, respectively. This validates the role of spatial attention in selectively emphasizing dental regions and suppressing less informative background areas, thereby improving localization and boundary precision.

Notably, both modules yield the largest improvements in AP75, underscoring their importance in enhancing segmentation accuracy under stricter overlap criteria. This indicates that fine-grained edge representations, facilitated by these components, are particularly crucial for high-precision instance segmentation.

4) Effect of Pyramid Branch Quantity: To investigate the impact of pyramid branch quantity on instance segmentation performance, we evaluate different configurations of the proposed Dual-Branch Pyramid, which is designed to enhance multi-scale feature fusion. Specifically, we use the framework incorporating only the Symmetry-Aware Module as the baseline. The variant with a single-branch feature pyramid is denoted as *Single-Branch*, while the complete dual-path version is denoted as *Dual-Branch*. The corresponding results are summarized in Table X.

TABLE X: Ablation study of pyramid branch quantity in the dual-branch pyramid.

Method	Tufts Dental Database		
	mAP	AP50	AP75
Baseline	59.3	90.7	**75.5**
Single-Branch	59.1	89.2	73.1
Dual-Branch	**59.5**	**91.8**	73.6

Compared to the baseline, the *Single-Branch* configuration results in a performance drop across all evaluation metrics, with declines of 0.2% in mAP and 1.5% in AP50. In contrast, the *Dual-Branch* setup achieves the best performance, showing

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improvements of 0.2% in mAP and 1.1% in AP50 over the baseline, and surpassing the *Single-Branch* model by 0.4% and 2.6% in mAP and AP50, respectively.

These results suggest that a single-scale pyramid is insufficient for capturing the diverse anatomical scales in panoramic dental images. Its limited receptive field aggregation restricts its ability to represent complex morphological variations and fine-grained boundaries. In contrast, the dual-branch architecture enhances the model's capacity to capture and integrate features from multiple spatial resolutions, enabling it to better handle overlapping teeth, ambiguous edges, and other challenging cases. This demonstrates the importance of robust multi-scale feature fusion for accurate dental instance segmentation.

VI. Conclusion

In this work, we presented a reliable and clinically oriented multi-stage framework for tooth instance segmentation and numbering in panoramic radiographs. The proposed system integrates three stages: supernumerary tooth classification, instance-level segmentation and FDI numbering, and missing tooth detection with post-hoc correction—to achieve high adaptability, completeness, and anatomical consistency. Central to our approach is the Symmetry-Aware Dual-branch Pyramid Network (SADP-Net), which explicitly models bilateral structure and scale variations through the SymAM and DBP modules. Extensive experiments across public and in-house datasets demonstrate that our method outperforms state-of-the-art baselines in both segmentation and numbering accuracy, especially under challenging anatomical conditions. The framework's modular design and strong generalization further highlight its potential for real-world clinical deployment. In future work, we will further refine the existing missing-tooth module so that it can be more seamlessly integrated into SADP-Net, enabling improved detection performance and computational efficiency while maintaining FDI completeness and clinical reliability. In addition, we will explore incorporating temporal dental records for longitudinal analysis, extending the system to pediatric and mixed dentitions, and integrating multimodal data such as intraoral scans to support broader clinical applications.

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This article has been accepted for publication in IEEE Journal of Biomedical and Health Informatics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JBHI.2025.3643805

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